

EXPERIMENT 2 — DATA HIDING USING XIAO STEGANOGRAPHY

Aim

To hide secret data from unauthorized users using XIAO steganography in order to maintain the confidentiality, integrity and availability of data.

Description

Steganography is the technique of hiding secret information inside another apparently normal file. The file containing the hidden information is called the stego file, while the original file used to hide the information is called the cover file.

Requirements

Category	Requirement
Software	XIAO Steganography
Files	A suitable cover file and a secret file
Security	Encryption method and password as required by the tool
System	Computer with the required software installed

PROCEDURE

A. ENCODING / DATA HIDING

1. Open XIAO Steganography.
2. Select or add the cover file that will be used to carry the secret information.
3. Select or add the secret file that has to be hidden.
4. Select the required encryption algorithm.
5. Enter a password for protecting the hidden information.
6. Start the encoding or embedding process.
7. Choose the destination and save the resulting stego file.
8. Verify that the stego file has been created successfully.

B. DECODING / DATA EXTRACTION

1. Open the steganography tool and select the previously created stego file.
2. Choose the extraction/decryption option.
3. Enter the correct password used during the encoding process.
4. Start the extraction process.
5. Select the destination where the recovered secret file should be saved.
6. Complete the extraction and verify the recovered file.
7. Open the extracted file, where applicable, to confirm that the hidden information has been recovered correctly.

RESULT ANALYSIS

XIAO steganography provides a method for hiding secret information inside a normal-looking cover file. The experiment demonstrates that confidential information can be embedded into a cover file and recovered later through the decoding process.

The experiment also demonstrates the use of encryption with steganography. A password is required during extraction, providing an additional security layer for the hidden information. Steganography hides the existence of the data, while encryption protects its contents.

The correct password and appropriate decoding process are required to recover the protected hidden information. The selected cover file must also be suitable for the operation, and changing the file after embedding may affect the hidden data.

RESULT

Successfully embedded a secret hidden file into a normal cover file using XIAO steganography with encryption and recovered the hidden data using the decoding process.

CONCLUSION

The experiment demonstrates how steganography and encryption can be combined to protect confidential information from unauthorised access.

OBSERVATIONS

1. XIAO can be used only with the file formats supported by the tool.
2. Not all files are suitable for hiding data because sufficient space is required.
3. The hidden data is embedded into a normal-looking cover file.
4. Changing the file after the embedding process may affect the hidden data.
5. The secret information is protected through the selected encryption method.
6. The correct password is required to recover protected hidden data.
7. A cover file is used as the hiding medium.
8. A secret file is selected as the information to be concealed.
9. The encoding process produces a stego file.
10. The stego file is used during the extraction process.
11. The extraction process requires the appropriate password.
12. The recovered file can be checked after extraction.
13. Encryption provides an additional layer of confidentiality.
14. Steganography conceals the presence of information.
15. The experiment demonstrates embedding and recovery of hidden information.

Practical Flow: Choose cover file → Choose secret file → Select encryption → Enter password → Embed → Save stego file → Select stego file → Enter password → Extract → Verify recovered file.